

VM Setup and Linux Kernel Compilation Guide

Advances in Operating System Design
Autumn 2026–2027

This guide covers setting up a virtual machine for the course and compiling a custom Linux kernel from source. All assignments in this course will use **Linux Kernel version 6.1.6**.

1 Virtual Machine Setup

1.1 macOS

- Use **UTM** as the virtualization software.
- Download the Ubuntu Desktop 24.04.4/26.04 LTS ISO from the official Ubuntu website.
- Create a new virtual machine in UTM and select the downloaded ISO as the boot image.
- Allocate a minimum of **50 GB** of storage to the virtual disk.
- Allocate adequate RAM and CPU cores to ensure smooth kernel compilation (recommended: at least 4 GB RAM and 2 CPU cores).
- Install **Ubuntu Desktop 24.04.4/26.04 LTS** inside the VM before proceeding to kernel setup.

1.2 Windows

- Use either **VMware Workstation Pro** or **Oracle VirtualBox** as the virtualization software.
- Download the Ubuntu Desktop 24.04.4/26.04 LTS ISO from the official Ubuntu website (24.04.4 is recommended).
- Create a new virtual machine in your chosen software and select the downloaded ISO as the boot image.
- Allocate a minimum of **50 GB** of storage to the virtual disk.
- Allocate adequate RAM and CPU cores to ensure smooth kernel compilation (recommended: at least 4 GB RAM and 2 CPU cores).
- Install **Ubuntu Desktop 24.04.4/26.04 LTS** inside the VM before proceeding to kernel setup.

2 Building the Linux Kernel

The following steps describe how to download, configure, compile, and install Linux kernel version 6.1.6 from source.

Step 1: Download the Kernel

Using `wget`:

```
wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.1.6.tar.xz
```

Step 2: Extract the Source

```
tar -xf linux-6.1.6.tar.xz
cd linux-6.1.6
```

Step 3: Install Build Dependencies

```
sudo apt update

sudo apt install -y \
    build-essential \
    bc \
    bison \
    flex \
    libssl-dev \
    libelf-dev \
    libncurses-dev \
    libncurses5-dev \
    dwarves \
    xz-utils
```

Step 4: Copy the Current Kernel Configuration

```
cp /boot/config-$(uname -r) .config
```

Step 5: Accept Defaults for New Kernel Options

```
make olddefconfig
```

Step 6: Ubuntu Configuration Fix

Ubuntu's `.config` contains Canonical certificate paths that do not exist in the vanilla kernel source tree. Open the configuration file:

```
nano .config
```

Change the following lines:

```
CONFIG_SYSTEM_TRUSTED_KEYS=""
CONFIG_SYSTEM_REVOCATION_KEYS=""
```

Step 7: Build the Kernel

```
make -j$(nproc)
```

Note: If you encounter compiler errors (such as strict type-qualifier warnings or issues with BPF tools) due to newer compiler versions in Ubuntu 26.04, you have two options. You can either start over using the **Ubuntu 24.04.4** ISO (recommended), OR clean the build directory, disable BTF generation and treating warnings as errors, and rebuild using the GNU11 C standard:

```
make clean
scripts/config --disable DEBUG_INFO_BTF
scripts/config --disable WERROR
make CC="gcc -std=gnu11" HOSTCC="gcc -std=gnu11" -j$(nproc)
```

After this completes successfully, continue to Step 8.

Step 8: Install Kernel Modules

```
sudo make modules_install
```

Step 9: Install the Kernel

```
sudo make install
```

This copies the kernel image and related files into `/boot`.

Step 10: Edit GRUB Configuration

```
sudo nano /etc/default/grub
```

Set:

```
GRUB_TIMEOUT=60
```

Comment out:

```
#GRUB_TIMEOUT_STYLE=hidden
```

Step 11: Regenerate GRUB

```
sudo update-grub
```

Step 12: Reboot

```
sudo reboot
```

Step 13: Select the New Kernel

When the GRUB menu appears, navigate to:

- Advanced options for Ubuntu
- Select **Linux 6.1.6** (not the recovery mode entry)

Step 14: Verify the Running Kernel

```
uname -r
```

Expected output:

```
6.1.6
```